

A Kantian Theodicy

When Hume awakened Kant from his dogmatic slumber, he fell out of the warm, comfortable bed of Wolffian rationalism into a bottomless abyss of skepticism from which he was never to escape. For the most part, he had no wish to and only hoped to save Newtonian physics and objective morality from skepticism's ravaging maw. This, in turn, he could do only by taking the subjective turn to idealism, putting the human mind in the place of God and thereby limiting our knowledge only to the objects of its own creation, thereby preserving the *a priori* necessity of scientific and moral knowledge at the cost of their substantiality. Newtonian physics is restricted solely to the realm of appearances, while morality, which owes both its content and its obligatory force to the operation of pure practical reason, seems to both invent itself whole cloth and to stand suspended in mid-air with no visible means of support. At least this seems to be Kant's intention and design.

In this latter case, however, Kant is not entirely successful, even by his own lights. He admits that morality ultimately needs a substantive grounding in something that lies beyond the all too limited resources for knowledge available to us on a strict reading of the *First Critique*, an invisible foundation beyond the grasp of any possible experience. Kant is able to provide only the flimsiest pretext for these required foundations, and few have thought his defense of them plausible.¹ I have no intention of attempting to extricate Kant from the well-known difficulties that afflict his moral theory in this regard. Nevertheless, within the ambit of a philosophy not limited by the Humean strictures accepted by Kant, the moral philosophy outlined in Kant's *Groundwork*

¹ However, I will attempt to provide such foundations in a forthcoming essay on the postulates of pure practical reason. Stay tuned.

contains many interesting and useful insights that can serve us admirably in a number of areas. One of these, in which Kant himself has anticipated us, is theodicy, the justification of the ways of God to man.

In this essay, I want to sketch the lineaments of a Kantian theodicy. Kant does not discuss the problem of evil as such in any of his published writings, although he did explicitly discuss it in his *Lectures on Philosophical Theology*, of which we possess only student notes.² However, there is much more that Kant has to say about this, in his various works, which can be of use to the theistic philosopher. To fully appreciate this fact and exploit it we need no less than a complete reconstruction of the practical philosophy outlined in the *Groundwork* - in outline, of course – supplemented by his other works on ethics and the philosophy of religion.

The Project of the *Groundwork*³

In the *Groundwork*, Kant's concern is to investigate the in principle possibility of morality as a going concern, something existing not as a mere possibility, but as something real and actual for us in this world. As such, this is no mere theoretical investigation. Instead, it is intended to be fully and completely practical, overridingly useful to us in living our everyday lives. To this end, we need to engage in a transcendental investigation of the possibility of morality so considered, one that reveals the indispensably necessary conditions for the existence of morality in the world as a going concern. From the very first, it seems evident that two such things are required: first,

² See the *Lectures on the Philosophical Theology*, Allan Wood and Gertrude M. Clark, Ithaca, NY, Cornell University Press, 1975, 115-121. Kant's discussion of the Problem of Evil bears a strong similarity to the "Irenaean" soul-building theodicy advocated by John Hick in *Evil and the God of Love*, revised edition, New York, Harper and Row, 1978. Surprisingly, Hick never mentions Kant's discussion of the problem of evil. It would be interesting to know what he thinks of it.

³ My interpretation of the *Groundwork* [*Grundlegung* - variously translated as "Foundations," "Fundamental Principles," "Grounding," etc.) is based primarily on two sources: Paul Dietrichson's lectures on this text at the UW, which I heard as his TA many years ago, and the writings of Allen Wood, which nevertheless I will cite sparingly.

the existence of a moral law and second, the existence of moral agents. Without the first, morality cannot exist at all, even in principle, since in that case there would be no action-guiding moral principles to govern the behavior of rational beings and thus no basis for imposing an ought on such beings in any case. However, the moral law by itself is not sufficient for morality to exist for us as a going concern. In addition, there must exist moral agents, beings capable of apprehending that law, recognizing its applicability to themselves, and conforming their behavior to that law. While most discussions of the *Groundwork* focus on Kant's derivation of the moral law, rather less attention is devoted to his account of moral agency understood as transcendently necessary for the actual existence of morality in the world as a going concern. In this paper, I hope to partially redress this imbalance by focusing mostly on the latter project.

From the very first, Kant insists that morality must be autonomous in such a way that its foundations and fundamental constitutive principles must be knowable *a priori* through reason alone, without relying on any disputable metaphysical claims or on merely contingent facts about the world and human nature. To this extent, then, Kant has to rely almost entirely on conceptual analysis in order to arrive at any firm conclusions about either the moral law or moral agency. This is a bold venture but not, on the face of it, a hopeful one and the great fascination of the *Groundwork* resides in how close Kant is able to get to completing it. Even if we judge that Kant has failed to achieve his main end his attempt is still a remarkable philosophical exercise from which much of value can be derived.

The first question, then, is the following: "What is necessary in order for there to be a moral law?" Taking the law of (physical) nature rather than statute law as his paradigm, and noting that the compelling force of such laws consists in their universality and lack of exceptions, Kant identifies the essence of law with the possession of these features. On this basis, Kant concludes

that, if there is to be any such thing as the moral law, there must exist universal and exceptionless normative principles to govern the behavior of rational beings. This, in turn, leads straightaway to the supreme principle of morality, which Kant calls *the* Categorical Imperative despite the fact that, on his account, every genuine normative principle takes the form of a categorical imperative. However, the derivation of the categorical imperative from the mere conceptual analysis of the concept of law is sufficient to establish the existence of the moral law, since the Categorical Imperative is self-exemplifying - an example of the very kind of principle it specifies as sufficient to constitute a moral law. If a universal, exceptionless moral principle exists, then so does the moral law. Since the Categorical Imperative is itself such a principle, this entails straightway that such a law exists. More than this, since the Categorical Imperative is itself a rule or law for determining contentful moral laws by the testing of concrete maxims, it also specifies the procedure by means of which concrete moral principles taking the form of categorical imperatives can be derived. Even if his derivation of the moral law proves inadequate - a topic I will not take up here - it is breathtaking that Kant can derive so much from so little with such great plausibility.

Kant's discussion of the nature of moral agency is much less rigorous than his derivation of the moral law from the mere concept of such a law and is spread throughout the *Groundwork*, but no less important for the overall success of his project. Kant does not specifically address the question "What are the transcendently necessary conditions for the existence of moral agents, beings capable of apprehending that law, recognizing its relevance to them, and conforming their behavior to its demands?" However, this question is never far from Kant's mind. The very first sentence of the *Groundwork* asserts that nothing in the world, or outside of it, has intrinsic value

except good will, something exemplified only by moral agents.⁴ This puts the moral agent at the very forefront of our understanding of morality. More than this, while the first two versions of the Categorical Imperative give us the form of the moral law, the third version provides us with the matter of moral law, bidding us to treat humanity, whether in ourselves or others, always as an end in itself and never as a means merely. This means that the primary subject matter of morality is the proper treatment of moral agents, whether ourselves or others. This also makes moral agents central to the entire moral project at stake in the *Groundwork*. There can be no morality unless there are moral agents; more than this, morality cannot exist as a going concern in any way relevant to us unless *we* are such agents. However, we cannot know whether or not this is the case until we have investigated the question of what a moral agent is. From the Kantian perspective, this once again requires a conceptual analysis of the notion of a moral agent, one that identifies the transcendently necessary conditions for any being to be count as a moral agent. I will proceed to this discussion in the next section, in which I will attempt to isolate these conditions, many of which are simply taken for granted by Kant or merely suggested by his remarks.

Transcendently Necessary Conditions for Moral Agency

The first and most obvious necessary condition for moral agency is that a moral agent must be a *self-conscious rational subject*. In order for morality to exist as a going concern, there have to be individuals capable of apprehending that law, seeing that law as applicable to their own behavior, and capable of conforming their behavior to that law. Since the moral law is a law or principle existing in the form of an unconditional imperative expressible in language, only a rational being could apprehend the moral law *as such* - as a normative/prescriptive, hence a *moral*,

⁴ In this essay, all references will be to the translation by James Ellington published under the title *Grounding for the Metaphysics of Morals* (3rd ed., Indianapolis, IN, Hackett, 1993), 7.

principle in the first place. Thus, unless there are rational beings, there are no moral agents, and morality cannot exist as a going concern. More than this, no being can be rational being unless it is also a self-conscious subject. Only such a subject can apprehend anything, let alone the moral law. A being is a *subject* if it is capable of, and actually enjoys, awareness of mental contents. All sentient beings are subjects in this sense since all possess a stream of consciousness consisting of sensations and complexes of sensations capable of eliciting a behavioral response. However, a self-conscious subject is one that, in addition to awareness of mental contents, is aware of those mental contents *as contents*, as that of which that subject is aware and hence as something other than the subject who is aware. Such a subject, then, apprehends/recognizes a distinction between that of which it is aware and itself as subject, as that which is aware, simply as something given to it in experience. The standard name for this subject is the *self*, the “I that accompanies all our representations.” However, a sustained examination of this notion will carry far beyond the essentially Humean account of the self presupposed by Kant in his discussion of the self in the *First Critique*.

Contrary to what Kant and subsequent German idealism teaches, the self is not apprehended by us as an object – there is no “phenomenal self” that exists as just one more representation to “inner sense.” As Hume points out, the self cannot be apprehended as a mental content or represented by any such content.⁵ Nor can it be the content of a “reflexive act” of awareness; as Sartre points out, the self retreats from and thus defeats any attempt to capture it from some “higher” viewpoint which, after all, it must itself occupy in order to accomplish this, thereby constituting itself as something other than what it is aware of as content through that

⁵ David Hume, *A Treatise of Human Nature*, L. A. Selby-Bigge, ed., Oxford, Clarendon Press, 1888, 251-263.

reflexive act.⁶ Instead, as I have argued elsewhere, the self apprehends itself simply through being itself and exercising the *per se* causality (in relation to the body) by means of which its activity of awareness is constituted. Since this *per se* causality is a sustaining rather than creative principle in its own right, it does not constitute the contents of which it is aware through its own activity - these have some other cause and so the self is “passive” with respect to them.⁷ At the same time, however, that there are any mental contents at all is still the consequence of the exercise of that activity of awareness by the self. While I may not be able to control what I see if I open my eyes, I can refuse to open them, in which case no visual mental contents will occur in my consciousness. We are aware of the self, then, as an active principle, exercising a simple, uniform activity in relation to its mental contents constituting them as such for me in sensation and perception and as performing operations over them in various kinds of mental acts, such as memory, imagination, judgment, belief, doubt, and so on. We thus have, as Berkeley would say, a *notion* rather than a representational *concept* of the self, since the self does not grasp its own essence by means of a representation in consciousness from which its substantial form can be abstracted. For this reason, as Aquinas says, the self knows itself in activity rather than through the apprehension of its own essence.⁸

The self as subject is also *substance*, as that which is self-identical in relation to time and as the transcendently necessary condition for temporal experience as something that actually

⁶ Jean Paul Sartre, *The Transcendence of the Ego*, Forrest Williams and Robert Kirkpatrick, trans., New York, Farrar, Straus, and Giroux, 1957, *passim*.

⁷ See my paper, “The Inescapable Self,” also on this website.

⁸ This, of course, is not Kant’s view, but as I have argued elsewhere Kant’s account of the self, like Hume’s is incoherent. My reasons for asserting this here are suggested *inter alia*. On the view of Aquinas, see Therese Scarpelli Cory, *Aquinas on Human Self Knowledge*, New York, Cambridge University Press, 2013. This book makes better sense of what Aquinas says on this topic than anything else I have read. Even so, Aquinas and other medieval Aristotelians undervalued and ignored consciousness in a way that it took Descartes to correct.

occurs in the world and absolutely inseparable from it.⁹ It is impossible for a set of mental contents, even if ordered successively in relation to “inner sense” (after the fashion of Hume) to constitute a stream of consciousness, precisely because those contents are such only in relation to a conscious subject that is itself not such a content and stands outside of that temporally-ordered series as the thing that is successively aware of those contents in experience. Without this, there is no “experience” as we enjoy it at all. Mental contents are not experients (my sensation of pain is not aware of itself, nor is my perception of redness self-perceiving) and so no amount of them concatenated together can constitute an experient, even momentarily. Quite the contrary, actually occurring experience requires that there be an unchanging subject of a changing, temporally ordered consciousness - a simple, sempiternal substance that exercises a simple, uniform activity that on the side of the subject has no temporal dimension but does so in relation to the changing substance in which that activity terminates. Consciousness, the interface between the substantial soul and the body, represents the body’s changing states, as these result from both its own processes and from interaction with external bodies, as a series of mental contents ordered in inner sense.¹⁰ Thus, without the substantial self there can be no apprehension of the moral law in this first place.

The self is necessary for morality in another sense as well. Only if the self remains identical in relation to temporal existence will it be possible for a moral agent to be subject to any sort of moral responsibility of the sort that Kant wants to attribute to moral agents. If I am not the same individual now as the individual who performed some action yesterday, but merely a Parfitean or

⁹ Both Hume and Kant reject this notion as well. Hume favors a “process” account of the self as merely a stream of consciousness constituted by externally related sense-qualities held together by the “mental gravity” of association. Although Kant does speak of the soul in a few places in the *First Critique*, and treats it as though it was a substance, such a claim is surely beyond what the Critical philosophy can justify.

¹⁰ I have explained this elsewhere in my “Body, Mind, Space and Time,” also on this website.

Nozickean successor to someone who existed yesterday, I can hardly be held responsible for what that person did, for I did not exist at that time and so cannot have that person's actions imputed to me as something done by me. In the same way, if I exist only at the present moment and will not exist tomorrow, then I cannot be under any moral obligation with regard to my future actions, since I will not exist at that time, nor will my successors or future selves be morally responsible or liable for any action that I do now. At most, I can be subject to moral obligation only at the moment that I act, a moment that quickly passes and puts me beyond any responsibility for my derelictions. Thus, neither in judgment nor in act am I subject to praise, blame, reward, or punishment for what I have done at any previous moment, hence for anything that I ever do at any moment. On these terms, morality is surely toothless.

Speaking of moral responsibility, Kant is notorious for having held that free will, understood as contracausal freedom or "liberty of indifference" is a necessary condition for moral agency. A moral agent is only such if it possesses the power of autonomy, the capacity to act in accordance with a rational, action-guiding principle solely because reason demands it and not on the basis of any non-rational, non-moral incentive for action. Not even awe and respect for the moral law can function as incentives for conformity to the moral law, says Kant. Their sole function is to "beat down," nullify, or mediatize the influence of non-moral incentives for action, which Kant collects under the general term, "self-love," by revealing them as worthless in comparison to the demands of morality. These feelings, rather than being the psychic motive for action simply create the psychic space in which the free, rational will can operate to directly determine our subsequent action solely on the basis of its apprehension of the moral law and that law's binding force without needing to rely on any extraneous sensuous motivation. The intuition

behind Kant's position here is captured in C. D. Broad's famous dictum, "ought implies can."¹¹ I can only be obliged to do that which it lies within my power to do, and only be held responsible for doing what I could have refrained from doing under the circumstances. However, if determinism is true, I can only do what forces outside of my control have causally determined me to do, so that it never lay within my power to have done anything other than what I did in fact do, and given that it was not in my power to have refrained from doing it, neither can I be blamed, praised, punished, or rewarded for having done whatever it was that I did.

It seems to me that Kant was correct about this. Liberty of indifference is sometimes parodied as the view that the free will lies suspended between alternatives for which there is no sufficient reason that determines my choice, with the result that my choice is the result of mere chance. However, belief in free will is fully consistent with the existence of various motives or incentives to act capable of intelligibly connecting that action with my personality and character, having various strengths, and so on. It only claims that these motives or incentives incline without necessitating and inform my action as reasons that prompt me to act rather than as necessitating causes of my "behavior." To treat such motives or incentives as necessitating causes makes them the product of forces outside my control, to whose operation I can neither contribute nor make any effective resistance. It alienates me from those motives and incentives, even my desires and reasons, by explaining their influence on me not in terms of their intrinsic qualities as desires and reasons for actions, but instead simply as coercive, irresistible causal impulses to behave in a certain way, impulses with a history to which I was unable to make any contribution and over which I was unable to exert any effective control. Such impulses may be operative in my

¹¹ Actually, what Broad said was "obligability implies substitutability," which is neither as catchy nor accessible as the more common "ought implies can." On this point, see Robert J. Richman, *God, Free Will, and Morality*, Dordrecht, Netherlands, D. Reidel, 1983, 20.

consciousness but they are in no way *mine*, because I have no ownership of them. Even if they influence me through my character and my reasoning processes, they do so in a way that makes them no more than the automatic working-out of forces already in play and that makes their intrinsic properties as preferences and reasons ultimately irrelevant to their causal force. Thus, if determinism is true, I never act on the basis of my preferences or my reasons due to the fact that I approve this preference or see the force of those reasons; rather, these are simply conduits through which ineluctable causal influence flows in such a way to determine my behavior.¹² My desires and reasons *as such* play no role in this process so that, in the ordinary sense of the term, we do not *act* (as opposed to merely behave) at all, and thus possess no rational agency whatever. In that case, neither can we possess moral agency.

Of course, Kant makes things notoriously difficult for himself as a consequence of his theoretical philosophy, which requires determinism to be true of the phenomenal world as a condition of the possibility of coherent experience. This, in turn, requires that the “empirical” self, i.e. the self that each of us is constituted as an object, be subject to complete causal determinism as well, considered as part of the phenomenal world. Since there is no room for free will as part of the phenomenal world, Kant is forced to postulate a second, “noumenal” self as the locus of free will. However, given the obscurity of Kant’s argument for phenomenal determinism and the incoherence of his notion of the noumenal self, we can be forgiven if we do not follow him on these points, even if this involves the rejection of his theoretical philosophy. If there is to be free will (and thus moral agency) at all, its *locus* must be the empirical self, which is the substantial,

¹² To put it a different way, I don’t do as reason bids me because I recognize the reasons as for doing so as true; rather, I judge those reasons as true simply as a byproduct of the operation of non-rational causal forces that have already determined what I will do *willy nilly*, independently of the truth of those reasons. In such case, acting on the basis of reasons is a mere illusion and action in accordance with true reasons merely accidental.

self-conscious, rational subject that each of us is, not merely a series of mental contents occurring in Kant's phenomenal world.

The foregoing considerations concern the transcendently necessary conditions for moral agency as such. However, these are not the only conditions that have to obtain in order for morality to exist as a going concern in such a way for it to occupy the central position that it does in Kant's account of the human person *qua* moral agent. For Kant, moral agency is the sole source of the intrinsic value of the human person, and thus of his or her inherent dignity and status as end-in-itself. Without this, the materially significant versions of the categorical imperative - the third, fourth, and fifth, would lack any significant scope or grounding. To put it another way, it is not enough for there to be morality that there should be free will; it is also required that there be significant moral choices existing as the object of free choice. To this end, two further requirements for morality as a going concern become evident to us: *vulnerability* and the *influence of non-moral motives*. Both of these are required if we are to have significant moral choices.

Significant moral choices will only be possible if the way in which we treat ourselves and others has a significant impact on us and our lives. Unless we are in a position to both cause significant benefits and permit significant evils in our own case and in that of others, we can have no significant moral obligations to ourselves and each other, and thus no important duties to speak of. If Martians, each of whom was encased in an impenetrable carapace and self-sufficient to supply all its own needs, were to come and live among us, we would lack any significant moral obligations to them due to the fact that it was impossible for us either to significantly benefit or harm them. There would be no form of treatment that we would be forbidden to impose on them, since they would be incapable of suffering or being harmed by it, and no treatment that we would be required to extend to them, since there would be no way in which that behavior could influence

them either for well or ill. Only if we were to discover some fact about them that made it possible for us to significantly impact their lives would we be able to conceive of any duty regarding our treatment of these creatures, even if they are rational beings. This is not the case, of course, where human beings are concerned. Our characters, welfare, and even our very existence, is ultimately dependent on how we treat ourselves and others. It is only due to this fact that the exercise of our moral agency is significant enough to count as something capable of constituting our intrinsic worth and dignity as persons.

At the same time, if the moral law were descriptive rather than prescriptive, such that we had what Kant calls a Holy Will (and which he attributes to God), once again we would not have any significant moral choices to make. Like the Leibnizian God, we would only be able to do what is best and would not be even so much as subject to temptation to do otherwise. In that case, we would exemplify the epitome of rational autonomy without, however, being able to exemplify Kantian good will in such a way as to possess moral worth. More is required for this than simply that we do the right thing for the right reason, i.e. because it's the right thing to do. We must also do so in circumstances in which dereliction from duty is a genuine possibility for us. This, in turn, will be possible only if we possess non-moral motives and incentives to action that can potentially conflict with the demands of morality. More than this, the possibility of significant moral choices will require that these non-moral motives arise in significant moral contexts and are felt with an urgency that makes dereliction from duty a genuine option for us in those circumstances. Otherwise, it will be *too easy* to be good, too easy to resist temptation, and thus no great achievement to do one's duty. No significant moral worth can accrue to us or to our actions in such a case.

In passing, it seems to me that it is in this context that we ought to understand Kant's doctrine of radical evil, which has recently been a point of focus in scholarly discussions of Kant's moral philosophy.¹³ Most students of the *Groundwork* are surprised to find that, despite his commitment to free will and to the ideology of the Enlightenment (which includes the rejection of the Christian doctrine of Original Sin), Kant does insist both that human nature *as such* has an ineluctable and ineliminable *propensity* to evil, one that expresses itself in an almost Hobbesian "unsociable sociability." While Kant declines to offer a "deduction" of this principle and merely waves his hand in the direction of everyday experience as proof of what he takes to be an obvious empirical truth, the foregoing gives us sufficient resources for offering such a proof from within the ambit of Kant's doctrine of moral agency.

If we are to have significant moral choices, then there have to be non-moral motives for behavior that are potentially in conflict with the categorical demands of morality. Kant collects all such motives under the name "self-love," though not all of these motives are selfish or even focused directly on the self and its wants. These non-moral motives are not evil in themselves, and Kant does not equate our propensity for radical evil with total depravity, explicitly denying that any human beings are governed by diabolical impulses that preclude a free choice for the good. Nevertheless, from the practical fact that self-love can, and so in many cases does, conflict with the demands of morality there is already in human nature *as such* a propensity to evil as part of the very possibility of significant moral choice. Since significant moral choice is necessary for the exemplification of any morally valuable level of good will in the world, every moral agent in the world has to experience, in concrete ways on numerous occasions, the conflict between what

¹³ See Sharon Anderson-Gold and Pablo Muchnik, *Kant's Anatomy of Evil*, Cambridge, Cambridge University Press, 2010. Allen Wood has discussed this in his books *Kant's Moral Theory* and *Kantian Ethics*.

morality requires on the one hand and what self-love motivates us to prefer and pursue on the other. In such cases, our rational apprehension of the promptings of self-love as being in opposition to the demands of morality constitutes them as spontaneously occurring (hence in that sense natural) inclinations to evil, resistance to which is necessary if we are to act autonomously and do what morality categorically demands of us.

Kant scholars have been puzzled by Kant's insistence that, despite being radical ("deeply rooted") in human nature that the propensity to evil is something for which we are responsible because we have chosen it somehow prior to making any actual moral choices. Kant may have made a misstep here; obviously, nothing can both the spontaneous prompting of ineluctable nature and yet also constituted as such by some "transcendental" free choice. What is clear is that Kant does not want to allow anyone the option of claiming that, since human nature is radically evil, that the evil we do is somehow irresistible and thus something for which we either lack personal responsibility altogether or with regard to which that responsibility is reduced or partially mitigated by that fact. The radical evil in human nature becomes part of our character through the exercise of free choice, by means of which the natural propensity to evil becomes an actual inclination to evil, one which because freely chosen makes the agent to be evil and thus liable to blame and punishment for his actions. Thus, one is not an evil person simply through possessing the natural propensity to evil, but only through freely acting on a maxim prompted by that propensity, which converts that propensity, the content of which is morally neutral considered in itself simply as an object of contemplation, into the morally evil *motive* informing my act. This conversion of the promptings of self-love from mere inclination to actual motive is due to its having been deliberately and intentionally adopted in preference to or abeyance of the demands of the moral law in circumstances in which I understand those demands, recognize their binding force, and yet

freely choose to defy, neglect, or ignore in order to do or get what I want instead. In so doing, I make myself evil through the misuse of my free will. I cannot blame my nature for this. More than this, my liability to blame and punishment is in no way mitigated, let alone abrogated, by the fact that it would not have been possible for me to have done that wrong without the presence of a spontaneous inclination that was not an object of choice for me, since I was free in the circumstances not to act on its prompting.

As I have emphasized, the promptings of self-love in the form of spontaneous inclinations are not intrinsically evil, but become inclinations to evil only due to an external condition, i.e. that in the circumstances to act as prompted by that inclination would violate the demands of morality. By the same token, it is arguable from the Kantian point of view that the presence of such inclinations to evil as objects of choice is not itself an evil. Significant moral choices, and thus the exemplification of good will, will be possible only if it is possible for us to choose contrary to the demands of morality, something which would not be possible without inclinations to evil powerful enough to make the object of that inclination a *genuine*, not merely theoretical, object of choice. Given that I have free will, however, it remains that on any occasion that I actually succumb to the promptings of an evil inclination I could have resisted it and freely chosen to act autonomously from my sense of duty. It is only my freely allowing this inclination to determine what I will do that is evil and this evil is something for which I am solely responsible. In turn, this moral dereliction constitutes me as an evil person through my own act of will, and with regard to which I am without excuse from the moral point of view. My every attempt to blame my nature, my circumstances, my parents, society, in short to justify myself and so deny my evil and my guilt, is

either self-deception or special pleading. Indeed, it is a further manifestation of evil in me, and thus an act no less wicked than it is common.¹⁴

Further reflection on the concept of moral agency as a necessary condition for the possibility of morality as a going concern (as opposed to something entertained merely hypothetically for purposes of conceptual investigation) would undoubtedly reveal other significant conditions and may even do so to profit. For my purposes here however, it will be sufficient to consider only the ones that I have thus far distinguished. My aim is to show that these considerations provide a straightforward and plausible justification of the “the ways of God to man” - a Kantian theodicy.

Natural Evil

Theologians since before the time of Augustine of Hippo have divided evil into two kinds: natural and moral evil. Natural evil is the ontic evil that results as a byproduct of non-rational processes operating in accordance with the morally neutral laws of nature governing the production of states-of-affairs in the world of material things.¹⁵ The operation of plate tectonics, for example, gives rise to volcanic eruptions, earthquakes, and tsunamis, events we classify as natural disasters because they often result in injury and death to human beings and widespread destruction to their cities and property. In a similar way, the forces of nature that govern the weather can produce severe events and conditions harmful to human beings: extended periods of extreme heat or cold,

¹⁴ I will not here discuss the controverted question concerning Kant’s proof of his claim that all human beings have in fact chosen to be evil nor that concerning how it is that evil be intelligible to us. For more on these topics, see Sharon Anderson-Gold and Pablo Muchnik, *Kant’s Anatomy of Evil*, Cambridge, Cambridge University Press, 2010, especially the articles by Muchnik, Allen Wood and Jeanine Grenberg, 116-194. Grenberg’s interpretation of Kant’s doctrine of radical evil comes closest to the view presented here. My own solution to the second problem can be found in my *How Free Will Works*, Eugene, OR, Wipf and Stock, 2011,

¹⁵ Ontic evil is the privation of the good that belongs to a thing by nature; ontic good is whatever answers to and is perfective of a thing according to its nature.

droughts, floods, hurricanes, tornadoes, thunderstorms, lightning, hail, forest fires, and so on. Various naturally occurring conditions can give rise to pandemic diseases such as plague, influenza, smallpox, tuberculosis, poliomyelitis, dysentery, diphtheria, cancer, and many others too numerous to mention. No human agency is directly responsible for the operation of these processes or the occurrence of these events and, although the more we understand these processes and their causes the better we are at predicting and their occurrence and coping with their aftermath, we lack the ability to altogether prevent them or contain their deleterious effects. At the same time, these events themselves are generally found to be the natural consequence of the operation of forces indispensable to the existence of life on this planet and whose overall effect is beneficial to its preservation. The ontic evils resulting from these events are thus, for the most part, merely byproducts of these events and processes and neither their purpose of nor the means through which the overall good ends of these processes are obtained.

In accordance with the traditional Principle of Licit Double Effect, it is morally permissible for an agent to perform an action with two consequences, one productive of ontic good and the other ontic evil, just so long as a.) the action itself is morally permissible, b.) the ontic evil produced is merely a byproduct of that action, rather than the means by which the ontic good is produced, and c.) the ontic evil produced does not outweigh the ontic good produced by that action. It is arguable that these conditions are met in the cases we have considered here. The natural disasters and pandemics that result from the operation of these forces and processes are intermittent, local, and generally of short duration, whereas the ordinary operation of these forces and processes that has these ontic evils as a byproduct is continuously productive of and indispensable for a much greater quantity of what are often more important ontic goods over an extended period of time. Given that even the most cleverly designed system of the world is still going to be imperfect and

flawed in some respects simply through being finite, and that the very existence and continued survival of those who would call the goodness of the creator into question on these grounds depends on those forces and their operation, it seems hardly unreasonable to judge that the Creator is not to be faulted for the occurrence of these ontic evils.¹⁶

However, even if it is not morally wrong for God to permit the bad effects of the operation of non-intentional natural forces, God is not like a finite moral agent, who has to stand by helplessly while the unintended bad consequences of his actions play themselves out in actuality. God, being omnipotent, has the power to prevent those bad side-effects, if only through miraculous intervention. As such, would we not expect such a Creator to *want* to prevent or mitigate these evils even if He is not morally required to do, simply out of love for His suffering creatures, even if not as a matter of moral duty? Indeed, one might even question the applicability of the PDE in this context; after all, *we* are obligated to prevent these evils and are excused from doing so only due to our inability - God cannot make the same excuse.

The sort of God that a Kantian might approve of has a ready defense against these misgivings. For Kant, the good will and its exemplification by moral agents is the only thing in the world, or out of it, which has intrinsic value. It follows, then, that the production and existence of moral agents is the sole and ultimate intrinsic good to be secured by the natural order of material things, which derives whatever value it may have from serving that supreme end. Further, as we have seen, our vulnerability, which is inseparable from our status as moral agents, is a necessary condition for the possibility of significant moral choices in this world. Given this, God is morally justified both in creating us in such a way that we are vulnerable to and dependent on the operation

¹⁶ Kant makes use of this principle to explain why God can allow moral dereliction without will it - see Lectures on Philosophical Theology, 117-118. Presumably, he could make the same move in this context as well.

of natural forces and not interfering (for the most part) in their operation in giving rise to ontic evil.

Our physical vulnerability is due primarily to the fact that we are embodied and indeed is inseparable from this fact. We can affect each other, for well or ill, only through the use of our bodies; as such, our vulnerability has to be subject to and derived from the operation of the laws of physical nature. It cannot thus be naturally limited solely to our interactions with other persons. Further, the fact that we are vulnerable to the operation of the forces of nature as well as to the behavior of other moral agents greatly increases our responsibility for ourselves and others, requiring that we not just refrain from harming ourselves and others through our own actions but also actively seek to protect ourselves and others from the dangers represented by the operation of those natural forces. Indeed, but for this the responsibility we have for ourselves and others would be greatly diminished, and the amount of good will we would exemplify would be greatly diminished as well. Nowhere is this more evident than in the case of natural disasters, where heroic responses to injury and suffering, both on the part of victims and those who aid them, are elicited and the full measure of moral heroism is evinced. Even where tragedy obtrudes and even our best efforts ineluctably fall short - a real possibility that must be present in order for there to be genuine urgency in our response, and being real, must and will sometimes be realized - Kant would say that good will would still shine forth like a priceless jewel even though it accomplishes nothing in that case. Moral heroism cannot be faulted for inculpable failure resulting from human limitations. As such, success in these cases needs to be possible (in order that our efforts not be pointless) but must not be antecedently assured in order to be productive of intrinsic moral value.

Given that God is morally permitted to allow these evils in the first place, and by allowing them makes possible substantial opportunities for the exemplification of good will on the part of

moral agents, the only intrinsically good thing in the world (or out of it) and one not dependent on consequences, He is also morally justified in not intervening to prevent the ontic evils that result from the operation of these natural forces. Instead, God brings moral good out of ontic evil, primarily through our moral efforts in working to mitigate, relieve and correct those moral evils, but also to provide warning, correction, chastisement, punishment, and reward to each in accordance with his or her spiritual condition and circumstances. More than this, for Kant what I have been calling ontic evils are for the most part so only in relation to the happiness that corresponds to self-love, not in relation to the moral agency from whose mere capacity our intrinsic worth as persons is derived. Due to this ontic goods and evils, regardless of their impact on happiness, are in no way comparable to the supreme value of good will. As such, happiness and moral worth are incommensurable and so cannot be compared quantitatively in such a way that we can judge that the amount, variety, or distribution of ontic goods and evils outweighs the intrinsic value represented by the exemplification of good will.¹⁷ These two kinds of good are *nonpareil* from the Kantian perspective and while it is true that there is an intuition that cosmic justice demands that moral worth (which is at the same time worthiness to be happy) ought to correlate with actual happiness, something that we do not see in this life, it does not follow that God is remiss in not securing it in this life. As is well known, Kant maintains that it is only in a future life that God will secure this end.

Nor is it inexplicable why this has to be the case. Our actions only possess moral worth to the extent that they are done from duty, not self-love. Thus, worthiness to be happy is a consequence of acting from duty as well and such actions will be possible for us only if a disparity

¹⁷ As, for example, does J. L. Mackie in his seminal paper, "Evil and Omnipotence," reprinted in Nelson Pike, ed., *God and Evil*, Englewood Cliffs, NJ, Prentice-Hall, 1964, 46-60, especially 53-59.

between worthiness to be happy and actual happiness are possible for us. This, in turn, requires that in acting from duty we are not assured of any actual happiness as an automatic reward for doing the right thing for the right reason, i.e. simply because it is the right thing to do. Only the fact that I am not assured of such a reward makes it possible for me to choose to do my duty regardless of its effect on my happiness and thus to exemplify good will, act in a morally worthy manner, and thus *deserve* happiness in the first place. Indeed, this will be possible only if there can be a real opposition between what morality requires and the demands of self-love, which in turn can only be the case if there is in fact such an opposition; for only in such cases can such a decision be demanded of and made by me. For this reason, we ought not to expect to see such a correlation in this life nor is God required to supply it here. Nevertheless, what is reasonably deferred for morally sufficient reason in the present ought not to be eternally denied, so we may reasonably suppose that a just God will right the scales, in another life if not in this.

It is important, however, that this be a matter of faith and trust in God rather than something that we can prove with the complete *rational* assurance. Otherwise, that assurance would once again rule out our choosing duty over self-love by guaranteeing an eventual reward that makes its self-interestedly rational to obey the moral law even in those cases in which I must make a temporary sacrifice of my self-love in this life. Thus even if, as Kant supposes, there can be neither rational proof nor disproof of the immortality of the soul, it is (as he also argues) at least *permissible* for us to believe that this is the case on moral grounds. Supposing this to be so, God will see to it that happiness and worthiness to be happy will coincide in the next life and that the demands of cosmic justice will be met. Without this, as Kant argues, the actual existence of the *summum bonum*, which is after all the *end* or *goal* of morality corresponding to its full and complete actualization in reality, will not be possible and may not even be approachable. In that

case, even in doing our best we would fall so laughably short of attaining it that we may well judge that to pursue it is a pyrrhic task, one that however mandatory from the point of view of autonomous reason is still hardly worth the effort and concern required to take it seriously as a (indeed *the*) central task of one's life.

From the point of view of a Kantian ethics, then, the theistic God is not to be faulted for creating a world in which natural evils exist, insofar as the ontic evils produced by natural processes operating in accordance with morally neutral laws of nature are mere byproducts of the operation of those forces, which are themselves indispensable for the very existence of moral agency in the world. Nor would a wise, providential God act in such a way as to systematically prevent or mitigate those ontic evils (though God might have good reason to miraculously intervene in the natural order on some occasions) since to do so would significantly reduce our level of responsibility for ourselves and others and thus prevent the exemplification of good will. Nor would it be prudent for God to insure that there was anything beyond a vague and general correlation between happiness and worthiness to be happy in this life, since that would make the autonomous choice for duty and against self-love impossible. Indeed, even the rational assurance of cosmic justice in the next life would mean that ultimately, in doing one's duty one is never really sacrificing one's self-interest or sacrificing self-love to do what is right, since one is assured that one is "laying up treasures for oneself in Heaven" and so guaranteed an ultimate payoff. Nevertheless, it is reasonable to believe and hope that the demands of cosmic justice will be met in the next life, given that God is good and just, both from the exigencies of that demand itself and as necessary for the realization of the *summum bonum* or at least its meaningful approximation by finite moral agents. A Kantian morality, then, sees no significant challenge to the existence or goodness of God arising from the fact of natural evil.

Moral Evil

Moral evil, by contrast to natural evil, is the ontic evil for which moral agents are responsible, either by mere negligence, omission, or deliberate refusal to do as duty requires. Kant emphasizes that, insofar as we are rationally autonomous individuals, we will always freely determine our actions in accordance with the demands of the moral law. To do otherwise is to allow ourselves and our actions to be determined by the promptings of non-rational motives and incentives for action. When we deliberately and intentionally do this in preference to the moral law, doing what we want to do rather than what we know we ought to do our actions, though the product of free choice, are positively irrational, i.e. contrary to the demands of reason and the moral law. Only moral agents, in fact, are capable of this kind of irrationality. Other animals, being non-rational, are incapable of moral agency and thus fail to meet the necessary conditions for exhibiting irrational (and therefore immoral) behavior.

As we have seen, from the Kantian point of view I can only choose to do my duty in such a way as to endow that action with moral worth if I do so freely, i.e. in circumstances in which it lay within my power to have done otherwise - minimally, to refrain from doing an act that I was otherwise prompted to do. This requires, however, that there be non-moral motives and incentives to act that potentially conflict with the demands of the rational moral law. Unless this is so, there will never be any significant moral choices possible for us. In turn, these non-moral motives must be non-rational ones as well, at least in the minimal sense that they are potentially opposed to the dictates of morality. It does not follow, however, that these non-rational motives are intrinsically morally evil or morally evil considered in themselves. Indeed, in non-moral contexts in which the demands of self-love have free reign, it is self-interestedly rational to act in accordance with the motives that maximize our happiness, understood simply as desire-satisfaction. However, when

we choose to pursue the ends put to us by self-love in abeyance or opposition to what the moral law requires, then *we* become morally evil as a consequence and so too do our actions through our free choice to pursue the ends of self-love in opposition to the demands of reason and morality.

God is therefore justified in endowing us with non-moral incentives and motives that are potentially in conflict with the demands of morality for the purpose of endowing us with significant moral choices and the opportunity to significantly exemplify good will. These motives and incentives are not evil considered in themselves and are not the causes of the evil actions done by evil moral agents - these are solely the result of the irrational, immoral *choices* made by moral agents who could have done otherwise. Of course, given that evil actions are a genuine possibility for us through the intentional misuse of our free will it, a consequence of this will be that evil actions will actually exist and occur through the dereliction of moral agents. Further, given the human vulnerability necessary for us to have significant moral choices, the consequences of those acts of dereliction will sometimes be gravely wrong and even tragic in their consequences, especially for the innocent. Yet, as we have seen, all of this is necessary in order for Kantian morality to exist as a going concern in the world, and for the supreme and sole intrinsic good, the exemplification of free will, to actually exist there. Since this is permissible in itself and the evil actions that result are solely the responsibility of finite moral agents, being merely a byproduct of the necessary conditions for the real possibility of morality as a going concern, in accordance with the Principle of Licit Double Effect God is once again absolved from any blame for permitting those actions.

Nor would it be wise or prudent for God to routinely intervene in the order of human affairs to prevent either the evil actions of human moral agents or their tragic consequences for others, even the innocent. Precisely because He is omnipotent and omniscient, if God were to take upon

Himself the task of preventing evil actions or their consequences, we would very quickly be robbed of the opportunity to make serious moral choices. People would soon discover that it was not possible for them to do succeed in doing evil actions and leave off trying to do so, since God would be able to anticipate and neutralize all such actions in such a way as prevent their bad consequences from obtaining. This would make it useless even to contemplate doing evil actions, leaving us with no alternative but to act in accordance with the moral law *willy nilly*. Conformity to the moral law in that case, being without alternative, would leave no scope for the exemplification of good will in such a manner as to endow our actions with moral worth thus preventing the sole intrinsic good, one that only morality can provide, from actually existing in the world. Instead of acting in this self-defeating manner, God works to bring good out of evil by commanding that finite moral agents work to prevent and eradicate evil in the world, a command that is not inconsistent with the possibility of evil actions actually occurring in the world and with people actually getting away with them but still making possible the exemplification of good will in the process of preventing many such actions from actually occurring and seeing to it that those that succeed in so doing receive appropriate temporal punishment for their actions.

Conclusion

The foregoing presents the main lineaments of a Kantian theodicy, one based on the characteristic teachings of Kant's practical philosophy. In this essay, I have specifically limited myself to the sort of philosophical theism typically discussed by philosophers of religion since the middle of the last century and have crafted a classical theodicy that does not rely on any

specifically religious or Christian doctrines. Despite this, I think that this represents a plausible response to the problem of evil as classically formulated, explaining how an omniscient, omnipotent, and omnibenevolent God would be morally justified in permitting the natural and moral evils we experience in this world. For those who understand the problem of evil in these terms, then, the foregoing should be recognizable as such a relevant response to their concerns and a basis for an ongoing discussion within the limits of that traditional conception of what the problem of evil is and what is required to solve it.

Of course, much more than this is required to deal with the problem of evil in its full dimension, for which it is impossible to completely eschew religious, and specifically Christian notions. In particular, the notions of sin, sinfulness, and the problem of evil itself as an expression of that sinfulness needs to be addressed. Here we will find that Kant is also a useful if unwitting guide, whose views on this topic are much closer to the Christianity he rejects than even he realizes.

However, that is a topic for another time.¹⁸**Imaginative Value Sensitive Design: Using Moral**

~~Imagination Theory to Inform Responsible Technology Design~~

Abstract

~~Safe-by-Design (SBD) frameworks for the development of emerging technologies have become an ever more popular means by which scholars argue that transformative emerging technologies can safely incorporate human values. One such popular SBD methodology is called Value Sensitive Design (VSD). A central tenet of this design methodology is to investigate stakeholder values and design those values into technologies during early stage research and development (R&D). To accomplish this, the VSD framework mandates that designers consult the philosophical and ethical literature to best determine how to weigh moral trade-offs. However, the VSD framework also concedes the universalism of moral values, particularly the values of freedom, autonomy, equality trust and privacy justice. This paper argues that the VSD methodology, particularly applied to nano-bio-info-cogno (NBIC) technologies, has an insufficient grounding for the determination of moral values. As such, an exploration of the value investigations of VSD are deconstructed to illustrate both its strengths and weaknesses. This paper also provides possible modalities for the strengthening of the VSD methodology, particularly through the application of moral~~

¹⁸ See my essays, "Sin and Suffering" and "Morality and Human Fallenness," also on this website.

imagination and how moral imagination exceeds the boundaries of moral intuitions in the development of novel technologies.

Keywords: value sensitive design, design psychology, moral imagination, applied ethics

1. Introduction

Developed for software engineering and systems development, Value Sensitive Design (VSD) was conceived by its founders Batya Friedman and others in the decade prior to the twentieth century (Friedman 1996). The methodology's intent begins with the premise that technology, and its design, is not value neutral; building on the ethical and political ladenness of artifacts within the field of science and technology studies (STS) which argues along similar threads (Pinch and Bijker 1987). The social and moral abstracts of design, particularly within that of the human computer interaction (HCI) community, are what have been of particular interest to Friedman and colleagues (Friedman, Kahn Jr., and Borning 2008; Borning and Muller 2012).

Beyond moral epistemology, VSD methodology assumes that upon close analysis of values, some may be determined to be universal among different cultures and societies. Although these values may manifest themselves in varying ways, upon close analysis one can decide that in fact, those manifestations are merely a variation of a universal value. This is the contention that is asserted by Friedman (Friedman 1997). As such, the goal of this paper is to argue that the universal values that VSD instantiates are built on contentious foundations. The contention is thus that, in light of recent developments in cognitive sciences, a revaluation of moral theory and epistemology is necessary if VSD is to remain effective across diverse societies and cultures. To this end, what follows is a thoroughgoing analysis of what the philosopher Mark Johnson calls 'moral imagination' and how Moral Imagination Theory (MIT) can reinforce the ethical investigations in VSD (Johnson 1993).

To the best of the author's knowledge, this paper is the first to explore VSD from a Moral Imagination Theory perspective, identifying the strengths and weaknesses of conceptual investigations that are central to the VSD framework in isolation in light of this perspective. Prior literature on VSD has focused on judging the applicability of moral values to technological systems (Brey, 2010; J. Davis & Nathan, 2015); VSD methodology (van den Hoven and Weekert 2008), applications of VSD to current technological innovations (Briggs & Thomas, 2015) as well as to novel technologies (Dechesne, Warnier, and van den Hoven 2013; Warnier, Dechesne, and Brazier 2014; Timmermans, Zhao, and van den Hoven 2011; van

Wynsberghe 2013; van Wynsberghe and Robbins 2014; van Wynsberghe 2016; Umbrello 2019b). Although all these studies provide useful information, they are built upon the existent assumptions regarding the validity of the moral values enrolled in VSD's value investigations. This paper's deconstruction of the enrolled moral values is comparatively unique; the intention being to spur continued discussions on some of the most pertinent ethical issues regarding the design of emerging and converging technologies.

To successfully tackle these arguments, this article is organized into the following sections: the first presents an in-depth account of the VSD framework. The goal of doing so is to lay out the methodology envisioned by VSD's founders as well as the framework's evolution to its current state. Particular attention is given to what VSD calls 'conceptual investigations', one of its tripartite constituents which investigates stakeholder values. As this is the part that explicitly focuses on, and is dedicated to, the exploration of moral values, it is of the most significance for this paper's analysis. The second section deconstructs conceptual investigations, mainly looking at the normative ethics which it takes for granted. In doing so, its normative foundations both independently and within the framework itself is evaluated. The final section of this paper provides a thorough overview of Moral Imagination Theory, why it crucial to conceptualize morality within this framework as well as how it can inform the VSD approach.

2. Value Sensitive Design (VSD)

VSD is a design methodology founded for technology design with the aim to incorporate human values⁴⁹ into early design phases in an encompassing and formalized manner (Friedman et al. 2013). Because VSD was birthed from the needs of informational and computational systems design, a design framework was necessary to encompass the value focuses of the field. Specific research emphasis within the field found that the values of privacy, ownership, property, physical welfare, freedom from bias, universal usability, autonomy, informed consent, and trust were each of substantial value to stakeholders to warrant a principled methodology for their incorporation into informational and computation systems (Friedman et al. 2013). As such, the VSD framework was developed in the effort to provide a principled way for designers to account for these values in early design phases (Friedman, Hendry, and Borning 2017). VSD has since been adopted as the potential design methodology for the applications of technologies outside of

⁴⁹ The recent literature in technology studies have argued for the inseparability and against the bifurcation of the nature/culture dichotomy, and similarly against the privileging of human values over the values of nonhuman animals (see for example Harman, 2018; Morton, 2016).

the information and communication technologies and HCI communities (e.g., Umbrello & De Bellis, 2018; van den Hoven, 2013).

Similar communities interested in value-based technological development have, and still are, in development and use such as participatory design, universal design and inclusive design. These different research communities, although focusing on the incorporation of values, differ from the VSD framework given that the former methodologies tend to prize instrumental and functional values such as user-friendliness (van den Hoven and Manders Huits 2009; Newell et al. 2011). VSD, on the other hand, aims to primarily investigate the stakeholder values of moral, rather than functional or pragmatic importance. These values most centrally include freedom, autonomy, equality, trust, and privacy justice (Friedman 1997). Although the instrumental values that may be enrolled in the design and development of the technology can be translated as values important to VSD, this framework's primary attention concerns moral values.

According to VSD founders, although rich with different meaning and definitions, values take on something that is beyond the factual distinction of which they may sometimes be equated. Thus, the fact/value separation is of critical importance here. One cannot be conflated with the other, and as such a fact does not logically entail that it is of value. Thus, what is of 'value' in VSD are those "interests and desires of human beings within a cultural milieu" (Friedman et al., 2013, 71), yet the reduction to universalism²⁰ and absolutism is nonetheless affirmed by VSD theorists and practitioners (e.g., Friedman & Kahn Jr, 2003; van Wynsberghe, 2012). Section 2 provides a more in-depth discussion of what values are of ethical importance and implicated in technology design. The framework, then is situated within the broad range of VSD literature from its founding values of import to more contemporary value investigations.

2.1 A Tripartite Framework

The design of technologies and the artifacts and techniques that emerge from their highly collaborative network of assemblages enroll a host of processes that form an implosive whole, that is it is greater than the sum of its parts (for a more in-depth analysis of this see Harman, 2016; Morton, 2018). VSD, in and

²⁰ Universalism is not equivocal with absolutism. The former argues that moral values are independent of socio-cultural differences, whereas the latter is concerned with intrinsically right/wrong actions.

of itself, is similar in this way too. VSD is iterative, it builds upon existing design practices and traditions as well as on itself as new information, techniques, issues, and advancements emerge throughout the design process. As such, VSD is composed of three distinct, yet necessarily interrelated parts or stages: conceptual investigations, empirical investigations and technical investigations.

VSD's method sets it apart from competing approaches because it is markedly self reflexive, falsifiable, and recursively self improving. Similarly, VSD is anticipatory and does not impute values or responsibility to individuals or groups solely on an ex post facto basis (van den Hoven 2017). VSD is thus predictive, aiming to ascertain potentially implicating values before they materialize and in response “influences the design of technology early in and throughout the design process” (Friedman, Kahn Jr., and Borning 2006, 12). What is outlined is only the first part of the methodology given that it is the one that is most philosophically based. The latter two, empirical and technical investigations, although feeding into the first, come after the stakeholders are delimited and values discerned.

2.1.1—Conceptual Investigations

This is considered the most theoretical among the tripartite investigations. Conceptual investigations are philosophically informed and require consultation with the philosophical literature. To this end, it aims to answer questions of limitations and constraints both prior to and throughout the design process such as: Who are the stakeholders? What are the situationally relevant values? Where are the boundaries when discussing often inversely relational values like convenience vs. safety? When do the limits of accepted practice come under question when conflicting with relevant values? What are the opportunity costs of one design over another? These questions and other theoretical areas are explored in conceptual investigations (see Denning, Kohno, and Levy 2013 who do this with smart homes). Research is thus undertaken that set the stage for the remaining two investigations, however, those proceeding investigations reflexively update the conceptual parameters that were initially considered.

3.—What Are Values?

Because it is the goal of this paper to articulate a novel framework from which VSD can conceptualize values, it is vital to articulate first what values are and how they are designed into artifacts. To begin, when entering into discussions about values, the lack of agreement between scholars and the existent literature makes any absolutist conception of ‘value’—whether ontic or epistemic—highly contentious

(Harmon 2016; Le Dantec, Poole, and Wyche 2009; Johnson 2014; Lakoff and Johnson 2003). To this end, this paper does not aim to argue for any such certainty, instead, this section will opt to discuss how VSD *per se* conceptualizes values.

In the VSD scholarship, a value is considered that attribute with which an individual or group consigns significance (see more specifically Friedman & Kahn Jr, 2003, 2). Throughout their discussion of values, Friedman & Kahn (2003) explore a genealogical account of values, and how, historically, they have been articulated. To this end, the authors' eventually opt for a deontic account of values that they believe can be distilled from a historical investigation of value conceptualization, particularly those that are of direct relevance to technological design and development (specifically Friedman & Kahn Jr., 2003, 6). The decision to formalize this set of values is due to their apparent trans-societal universality. However, the authors do recognize the incommensurability of these values, their dynamic character depending on what technology is under consideration and the socio-cultural context in which the design is being undertaken. Still, the underlying characteristic of these chosen values, despite their varying relevance, are taken as universal regardless of the socio-cultural hermeneutics that is employed to distill them.

With this in mind, how are values—at least in the VSD sense—designed into artifacts? Technology philosopher Philip Brey provides a salient way of conceptualizing this integration:

The idea of embedded values is best understood as a claim that technological artifacts (and in particular computer systems and software) have built-in tendencies to promote or demote the realization of particular values. Defined in this way, a built-in value is a special sort of built-in consequence...Then tendencies for the promotion of values are identified as special kinds of built-in consequences of technological artifacts (Brey, 2012, 3).

Because of this, values are instantiated and expressed via their use. To this end, some values that may not have been manifest *prima facie* may express themselves when implemented by users. Dangers that may result are easy to envision, particularly when cognitive biases are implicated in technological design (Caviola et al. 2014). Although some work has been undertaken to limit the effects of cognitive biases in technological design, the danger that implicit or biased values may be embedded lingers nonetheless and calls for a more holistic conceptualization of values that is in line with developments in the cognitive sciences and their findings on the basic structure of human cognition and moral reasoning (an ad hoc functionalist means of de-biasing VSD has already been undertaken, see Umbrello, 2018b).

For this reason, it is argued that moral imagination provides an effective framework for conceptualizing the 'design turn in applied ethics', and in particular VSD. It is argued that the current account of VSD is predicated on an insufficient account of what constitutes values and moral deliberation. Moral imagination is argued to be a more genuine account of how humans engage in moral theorizing and this paper proposed that it should be incorporated into VSD should it aim to be effective. The following section presents a thorough layout of what MIT is, why it is preferable over the Enlightenment tradition of moral theorizing as well as how VSD can be modified to better account for values in design.

4. Moral Imagination and Imagining Possible Futures

From Plato onwards, the continued variance between the existent theories of morality is, in itself, an indicator of the often conflicting commitments, values and aims that individuals find themselves in their day to day lives. Ultimately, distilling any of this 'moral overload' (see van den Hoven et al., 2012) to a single, universal or absolutist principle for grading diverging commitments falls short; something is always sacrificed, albeit even if intuitively, when adopting any moral law tradition (e.g., Stocker 1976a).

This process calls for, as Mark Johnson indicates, "moral imagination...in all its various manifestations, as a means to both knowledge and criticisms" (Johnson, 1993, 187). What this means is a reflexive understanding of the self and the imaginative structure of moral deliberation including its principles and constraints. This also applies to those who reside and ascribe to traditions of morality that differ from our own. Imagining available possibilities that can translate, inform and modify our behaviours and actions and how those imaginaries affect our moral deliberations is what is involved. Moral growth is thus attained via an *imaginative rationality* aimed towards the understanding of cultures, institutions, systems and the self (Johnson 1993).

Moral Law Theories (MLTs) such as deontology, utilitarianism and their various offshoots rarely, if at all, result in moral growth. What they do is instead argue, mistakenly, that there are a constrained set of rules, with definitive actions for making moral decisions. This is typically undertaken in prototypical cases that are themselves constrained and unambiguous. Instead, how they should be viewed is as a distillation of a common set of past collective practices, situated within a particular moral habitat that applies to unproblematic cases. However, these MLTs become ineffable when confronted with the moral overload of day to day life and even more so with morally ambiguous technologies, of which no rooted rules or laws can clearly resolve (e.g., Caviola et al., 2014).

Moving beyond resolute theories on how to behave, or that they *should* give us rules is difficult, and particularly contentious within the analytic tradition. To this end, to formulate a strong argument for why MLTs, as they are envisioned within safe-by-design (SBD) approaches to technological development, this section outlines MIT. Naturally, given the length constraints of this paper, the account detailed here will be skeletal, and the lack of multiple examples, case studies, and narrative modalities should not be construed as an argument against the strength of MIT. Readers should make careful note of the full-length arguments existent within the scholarship that adequately buttress moral imagination despite its current lack of popularity and dissemination (Johnson 2014; Fesmire 2003).

4.1 Situating Moral Imagination

The crux of MIT is founded on the consideration that moral reasoning is fundamentally imaginative and is constituted, given advances in the cognitive sciences, by shared essential imaginative elements. Here a brief account of each of these proposed elements is given.

4.1.1—*Concepts as being Prototypical*

The classical understanding of basic concepts as being accountable through a sufficient/necessary framework does not take into consideration that even the most foundational notions of cognition have an architecture that is not exhaustible by this traditional account of concepts. For example, the concept of *love* is prototypical in the sense that there are undoubtedly categorical constituents that are paramount—both cognitively and experientially—but there are also peripheral constituents which, although not central to the concept itself, are more fundamental to critical examples.

To this end, Johnson (1993) argues that MLTs must be abandoned if, in fact, our basic moral concepts (e.g., harm, love, justice, etc.) are prototypical in constitution. This is based on MLTs erroneous account of concepts and their resulted view of moral judgments founded on the concept understanding landscape. The ways that MLTs seeks to apply absolute judgments²⁴ to samples by designating features via a

²⁴ Here ‘absolute’ refers to ‘moral absolutism’ in which there is an objective basis on which ethical beliefs can be evaluated.

sufficient/necessary approach is itself insufficient; many of these concepts are not ontic in the real world as MLTs characterize them.

This does not mean that MLTs are not functional. Prototypical cases can, usually, be accounted by MLTs without much controversy within a given moral framework. However, the existence of these very moral principles is predicated on their ability to address the prototypical, fundamental constituents of moral concepts, rather than their peripheries.

In contrast, quotidian moral issues surround, instead, the aprototypical situations that individuals find themselves in. However, it is precisely located here, on the edges of those prototypical concepts where the rules prescribed by MLTs fail to function. This is because MLTs cannot, in and of themselves, be extended to encompass these types of cases. Why? Because MLTs are themselves constituted by strict rule judgments that do not include the capacity for imaginative amplification. Nonetheless, it is this very imaginative ability to address day-to-day moral concerns that is necessary, something MLTs are not equipped, and never were equipped, to provide.

Still, we do possess the capacity for making informed moral judgments. Precepts that allow for application from central prototype cases to peripheral aprototypical cases exist. Metaphor is perhaps one of the most potent tools available to individuals for making this center-periphery leap. Our socio-cultural embodied experience is what permits us to attribute differentiated levels of significance to some instances rather than others. We learn about love, not as a formalized set of features/rules that constitute the central concept, but from prototypical experiences of dating, spatiotemporal closeness (and separation) as well as physical affection; however, those prototypes are not exhausted by these experiences (Lakoff and Johnson 2003).

Similarly, the experience of these prototypes is situationally contextual, and this situatedness comprises dynamic and incommensurable affect. This means that various emotional and cognitive states can emerge as a result of this situatedness, all of which are different catalysts for action. To this end, the essential moral concepts that we possess cannot be reduced to disembodied, purely rationalistic features, but are fundamentally co-constituted by our varied emotional states of embodiment. This means that prototypical models are necessarily dynamic and account for various value changes. Challenging, often multi-faceted situations that emerge in continually changing societies require greater encompassing power that move far beyond the more primordial prototype from which they began. The co-construction of the

prototype—aprototype amplification means that the features of central prototypes will similarly be modified across time.

In sum, because of MLTs' inability to account for non-abstracted contextuality, their potency is contingent on their applicability to the prototype itself; MLTs cannot remain potent independently. Similarly, moral growth requires the extension of prototypes to aprototypical cases via imagination amplification. This results in a cathartic dynamism of prototypes and is precisely what permits moral flourishing.

4.1.2—Prejudgments

Like contextualization described in the preceding subsection, the manner in which various cases are conceptually confronted is of critical import; like metaphors and prototypes which are central to the way individuals approach and conceive circumstances. The way these confronted circumstances are approached and understood is fundamental to the actions taken (or not taken) in response. Consider an imagined case where a terminally ill individual is given experimental nanorobotics to deliver targeted pharmaceuticals without consent and dies as a result. Conceived as an unethical application of power on behalf of medical professionals, we may respond in one way, however conceptualizing it as an empathetic, last ditch attempt by doctors to save the already dying patient's life may result in a very different response.

Language and semiotic structures often implicate imagination, mostly through the enrollment of prototypes and metaphors, among others. What these structures do then is not reflect some indisputable, external concept/case, but translate it. Because the prejudgments that we bring with us into situations where moral evaluation and action is undertaken changes the way we act, a reflexive self analysis of what prejudgments we are in possession of is necessary if we wish to arrive at an authentic understanding that human moral judgment is contingent upon.

4.1.3—Metaphor

This paper has so far consistently brought the concept of metaphor into accounting for what MIT consists of, and because of its central importance to the theory it warrants further detail.

The basic contention of Moral Imagination Theory is that moral deliberations hinge on the use of at least more than one metaphor and not the universal application of moral laws onto external cases (Johnson 2014). This is a profound shift away from traditional conceptions of moral epistemology and the sources of moral knowledge. Novel issues emerge out of this metaphorical extension and conceptualization of moral epistemology that could not have been devised by MLTs because metaphor has no place within those frameworks of morality at all.

MLT, and post-Enlightenment Western philosophy, have typically conceived of metaphor as lacking any rational basis to be considered part of moral epistemology (i.e., the reason/desire bifurcation) (Wood 1999; Borges 2004).²² The cognitive/emotive dichotomy is typically asserted as a means to avoid the claim that metaphor is implicated in situational reasoning. Similarly, dissent regarding metaphor comes in another form, that being that they lack any objective determinacy to be sufficient for moral knowledge. As such, their content cannot map sufficiently onto cases nor give clear direction on correct action. The precarity that is often associated with extreme moral relativism is cited when conceptions of the metaphoric nature of human cognition, particularly in moral understanding, is discussed.

However, the fear of extreme relativism can be averted once there is a genuine understanding of metaphor as a universal process of comprehension, one that is curtailed by common biological, sociocultural, and institutional effects. What must be acknowledged, however, is that a plurality of possible actions can be taken in any situation, none of which are the deduced offspring of univocal universal laws. What MIT is then is a very restricted and limited form of moral relativism (Johnson 1993).

Here it is useful to cite one of the specific examples that Johnson uses to explain the metaphoric nature of cognition. He identifies a set of particularly salient moral metaphors that are morally significant. One such metaphor is what he calls the *moral accounting* metaphor that aims to account for the increase/decrease of well-being. He divides this metaphor into five possible instantiations/interpretations of which he calls 'schema'. Due to space limitations, only one here is listed solely as an illustration:

Schema 1: RECIPROCATION: 'one good turn deserves another'

²² See also Critique of Pure Reason where Kant states along these lines that; "Inclination is blind and servile, whether it is kindly or not; and when morality is in question, reason must not play the part of mere guardian to inclination but, disregarding it altogether, must attend solely to its own interest as pure practical reason" (Kant, 2015, 5:118; see also KpV 5:71–72)

EVENT: A gives *something good*²³ to B.

JUDGEMENT: B owes *something good* to A.

EXPECTATION: B should give *something good* to A.

MORAL INFERENCES: B has an obligation to give *something good* to A. A has a right to receive *something good* from B.

MONETARY INFERENCE: B pays A for getting *something good* by giving *something good* (of equal price).

EXAMPLES: I owe you a favour for that good deed.

————— You saved my life! How could I ever *repay* you?

————— You've been so kind; I'm deeply *indebted* to you.

————— Your generous acts have *earned* you my respect.

I owe you more than you'll ever know, for what you've done for me.

She *bought* his respect with her constant good will. (Johnson, 1993, 47.)

What is happening here is a linguistic illustration of the structure of our moral reasoning, one of the innumerable examples within Western culture of how we speak and reason about morality (Taub 2012; Ervas, Gola, and Rossi 2018). The cognitive sciences have pushed beyond the inherited Enlightenment tradition of faculty psychology regarding the strict bifurcation of reason/desire resulting in a radical re-envisioning of the structure of human cognition. Examples of this are the concepts of *genetically transmitted response strategies* and the *epigenetic rules* of sociobiology (De Tiège et al. 2018; Segerstrale 2016). The once thought natural split between the two selves (in itself a profoundly theological construct) is reformulated as co-constituting one another, intermingled in constant evaluation and self-analysis that makes their a priori separation a fruitless, and philosophically erroneous endeavour (see Kahn, 1987; Filmouth, 2007).

In reevaluating the nature of moral knowledge, primarily by acknowledging its metaphorical structure, Johnson lists some of the implications for moral understanding that follow:

- Because our moral understanding is metaphorical, understanding the nuances and specifics of that structure is essential to arriving at a more general understanding of self. In doing so, we can arrive at an understanding of our own prejudgments and similar values (Johnson 2013).
- Perhaps the most important implication of metaphoric evaluation is that investigating metaphor allows us to determine potential contenders for univocal morals. Cross-culture experimental investigations can be undertaken to determine if specific metaphors, like the *moral accounting*

²³ Here 'something good' is defined intra-metaphor as a "valuable commodity (or its monetary equivalent)..." (Johnson, 1993, 47).

metaphor, are essential to human practice. Methods have already been outlined for how this can be undertaken (Gibbs Jr 2017).

- The universal understanding of the metaphoric structure of cognition allows us to ascertain, more specifically, where sociocultural influences come into play (such as the concept of transperspectivity outlined by Putnam, 1981, particularly at 168). Still, although similar metaphoric structures may be universally grounded, their instantiations and definition of values may nonetheless be markedly varied.
- A general understanding of the metaphoric structure of cognition may show that the specific metaphors adopted by individuals, those who define their value structures, serve as the nail in the coffin for any ontological status of absolute moral values. What may constitute 'absolute' in this sense, general or not, may be so divorced from its individual instantiation that its employment as a moral law would be absurd (Johnson 2014). However, the sociocultural instantiation of any particular metaphor may be so potent, despite being domain specific, that it directs the actions within the group to a determinate degree (Goldberg 2016).
- An emergent issue determines how well established is a metaphor, or network of metaphors (for more on embeddedness see Fauconnier and Turner, 2008a, 2008b). Naturally, the more central, or essential a metaphor is, the more distributed it will be as a node in a network of other metaphors. The essential nature of a metaphor is of concern, especially when discourses of change and critique come into play. The more established a metaphor is in cognition, the lower is its ability to shift. This does not entail the impossibility of change, but such change often comes with devastating sociocultural expenditures; i.e., European/North American colonialism is a good example of change viz. coercion.
- Aprototypical cases continually present themselves within the dynamic experience of being. Prototypical concepts must then be expanded to confront these. Metaphor is one of the cognitive structures that is employed in making this extension. Many of the moral issues that we confront when exposed to an original case is not the case *per se* but the difficulty in mapping prototypes on to aprototypical entities. Technology, in particular, presents continually dynamic and novel cases that stretch our moral prototypes in ways that enlightenment structures of moral deliberations simply did not, and which could not be accounted for essentially (Ervas, Gola, and Rossi 2018).
- Moral growth is gained by metaphorical extrapolation from similar, yet different, past experiences to new ones. Metaphorical accommodation/assimilation can only be possible because of the plasticity of metaphors.
- One of the critical points about the ontological nature of metaphors is that we can only have limited epistemic access to them. Because they are entities/objects that are consistently

networked, interconnected, and enmeshed with other metaphors and cognitive structures, total understanding is impossible and any attempt to exhaust a metaphor through literal or relational strategies fails to capture it sufficiently (for a further exploration of the 'withdrawn' nature of objects/entities see Harman 2018a). One of the primary failures of the analytic moral tradition is its ultimately failed attempt to exhaust moral identity into a rule-based system that sidelines the very foundations that give this identity value.

In sum, moral epistemology hinges on an intimate comprehension of how metaphors structure and are continually structured by human moral thought and action. In doing so, the varied—although potentially universally based—instantiations of moral metaphors force us to accept that cases of moral overload have a plurality of potential ways of being addressed, rather than a single, universal rule. This does not mean that radical subjectivism/relativism are affirmed, as a consequence (that would be a false dilemma). Biopsychosocial restrictions strongly delimit the contentiousness of subjectivism by limiting the foundations of specific metaphors within cultural bounds (e.g., like the psychobiological response patterns, see Kemeny, 2009). The latter is avoided because of the cognitive sciences which have shown the universality of some metaphors as being the consequence of embodied being (Varela, Thompson, and Rosch 2017; Gibbs Jr 2006).

4.1.4—*Narrative*

The function of the narrative structure of moral understanding and growth is often overlooked within the analytic tradition of moral theorizing. It has been noted that individuals seeking moral growth turn towards narratives (in the physical or performed modes) over the formulaic moral texts of philosophers (Rorty 1989; Peterson 1999; Mordini 2007).

The ultimate motivations for why narratives have such moral importance, and thus impact on moral growth lie in the narrative structure of day-to-day human *being*. Our embodied, lived experiences within a structured narrative permit agents to make discrete discernments between what is of moral import or not in any given situation (Krakowiak and Tsay-Vogel 2015; Kramp 2003). What we do when we engage with performed narrative (books, plays, etc.) is that we engage in a vicarious projection of ourselves and embody the character's thoughts, actions and moral anxiety (i.e., Sarbin 2001). In doing so, we engage in a reflexive dialogue and critique of their chosen paths and their actions which, as a function of our very

engagement in this behavior, impacts and develops our own moral understanding (Haidt and Joseph 2007).

Although the empirical literature on the narrative structure of moral growth (and similar cognitive modules) has been the product of the cognitive sciences, there has been philosophical work grounding the value of a narrative understanding of morality in philosophy as well. The vanguard of this tact has been by Nussbaum in her critical exploration of classical Greek tragedy, *Fragility and Goodness*, where she perceptibly states:

a whole tragic drama, unlike a schematic philosophical example making use of a similar story, is capable of tracing the history of a complex pattern of deliberation, showing its roots in a way of life and looking forward to its consequences in that life. As it does all of this, it lays open to view the complexity, the indeterminacy, the sheer difficulty of actual human deliberation....*A tragedy does not display the dilemmas of its characters as pre-articulated; it shows them searching for the morally salient; and it forces us, as interpreters, to be similarly active* (Nussbaum, 2001, 14, *emphasis mine*).

Narrative, like life then, is not so determinate as the arguments moral philosophers are so apt to construe. Like Kant's impenetrable *das Ding an sich*, no amount of re-contextualizing and forceful determinacy can exhaust the nuances and contextual contingencies of any given situation that an individual can, and will, inevitably find themselves. What narrative does then, at the very least, is bring a genuine colour to the portrait of which the rules of moral philosophers have only captured a tiny, but still important outline.

Narrative, to this end, allows for a more holistic and situated evaluation of our diverse moral landscapes *in time*. Our various, and sometimes conflicting moral enmeshments can be engaged within a continually developing, rather than ad hoc, way (van den Hoven, Lokhorst, and van de Poel 2012). Narratives are a mode towards enrichment that cannot be captured in any meaningful sense, nor exhausted by moral laws (for more on exhaustion see Harman, 2016, 2018b).

Even when we consider moral laws in themselves, their force—if that force becomes manifest—only emerge through a collective and experiential understanding of how these laws are embodied and function over time. Their salience is given force through narrative, not viz. some independent objectivity (Haraway 2016).

In sum, the narrative structure of human experience and cognition must be given its due role in moral philosophy. This becomes particularly apparent as the continued development of transformative²⁴ technologies shifts socio-cultural, political and economic norms. Any sufficiently potent anticipatory design approach adopted for the development of these technologies must embody an imaginative architecture that more authentically maps our moral cognitive structures to our situated realities.

5. Assembling a *Dynamic Value Sensitive Design*

This project thus far has demonstrated two things: 1) it has given an outline of the motivations for VSD's (and other SBD approaches) emergence as well as a brief account of its structure and methodology; and 2) it has given a cursory argument of and for why MIT theory maps on to human reality in ways that MLTs are simply not equipped to do. The primary motivation behind this is that VSD and similar methodologies aim to address the value-ladenness of technology design through value investigations that essentially draw from the existent philosophical literature. The dominant analytic modes of moral inquiry have been the traditional sources of philosophical interest, but the 'ecological turn' in science and technology studies has resulted in a greater tapping of the continental tradition, mostly at the intersection of cultural anthropology (Morton 2018; Latour 2017).

Still, much of the VSD literature has, and is currently, drawing from the inherited Enlightenment tradition based on faculty psychology, primarily the deeply anthropocentric post-Kantian tradition of correlationism²⁵. As this paper has argued, this is untenable, mostly because the moral laws distilled from these traditions are not reflective of the constitution of human cognition, primarily its failure to account for the imaginative structures such as metonymy, metaphor and narrative that have been shown as essential to moral deliberation and development. The issues are exacerbated with the introduction and co-construction of transformative technologies, primarily NBIC artifacts that push the limits of our moral intuitions (Umbrello, 2018b) and stretch any embedded deontic ethics to inoperability beyond any strictly delimited and prototypical cases.

²⁴ Transformative being the operative term here, it's particularly within a narrative and linguistic evaluation where substantial change occurs.

²⁵ Correlationism was first coined by Quentin Meillassoux (2008) to describe the post-Kantian anthropocentric conceit that "we only ever have access to the correlation between thinking and being, and never to either term considered apart from the other" (Meillassoux, 2009, 5). This idea has been a common ground for criticism that has united the speculative realist philosophers.

Autonomous vehicles (AVs) prove to be a timely example, and a predictive precursor to more advanced autonomous technologies, particularly in the socio-ethical issues that emerge with their introduction (Etzioni and Etzioni 2016). Primary concerns with the ubiquitous rollout of AVs are when they are confronted with trolley-like ‘dilemmas’ of life and death as well as what designers and programmers can do to address these issues (Gogoll and Müller 2017). Primary recourse has been to ad hoc implementations of hardware that can transfer the action potential and consequential responsibility onto the driver (Contissa, Lagioia, and Sartor 2017). A similar strategy has been taken to offset the inevitable inclusion of cognitive biases when enrolling stakeholders to extract values for design, mostly by applying de-biasing heuristic tools (Umbrello 2018). Still, these approaches to design are severely restricted, mostly because it requires the programming of prototypical cases and strict responses. The embedding of MLT approaches to decision making inherently leaves some degree of moral overload, meaning that the option that the programmers embed do not exhaust the moral tension that the proposed ‘dilemma’ appear to evoke. This is not to say that the MLT approach does not have some functional value, the reality remains that these technologies are being designed, rolled out, and integrated within the socio-cultural milieu in which they are introduced. Ethical issues arise and must be addressed, this is an excellent preliminary avenue that can be taken, but further conceptual, technical, and empirical research must be undertaken for future technology assessment.

5.1 Shifting Focus Towards Designers

One of the most substantial progressions in applied technology ethics is one that emerged as the backbone of the *design turn*, primarily the shift of focus away from the artifacts themselves and towards the activities and values of stakeholders (both direct stakeholder such as designers and indirect stakeholders such as publics). Still, substantial emphasis in technology assessment and speculative techno-ethics discourses is placed on the moral agency of technologies per se. This is seen more often than naught in the literature on artificial intelligence, primarily artificial general intelligence (AGI) and artificial superintelligence (ASI). Although there is considerable value in speculative ethics (see especially Roache, 2008), care must be taken not to commensurate current forms of AI and those of a futuristic, although arguably inevitable forms of AI (Müller and Bostrom 2016). The operative point here is that the scholarship has argued that the latter (AGI/ASI) constitute moral agents and similarly initiates discussions on personhood and rights attribution; the former, however, do not (Rolf and Crook 2016; Hughes 2014).

To this end, designers and ethicists have, and still are, considering which ‘human values’ to incorporate in design, how to align technologies—AI for example—towards pathways that are beneficial for human

civilization (Umbrello and Baum 2018; Baum 2016). Aside from the blatant anthropocentrism and dominant Western moral ideology that are usually uncritically assumed in these discussions²⁶, particularly when totalizing conceptions of ‘the world’ and ‘human civilization’ are invoked, the modern deontic theories of morality (utilitarianism and deontology) are usually assumed as the philosophical solution, shifting the issues and attention towards technical means to operationalize them.

What is needed then is a fundamental re-evaluation of the strategies adopted by designers and ethicists in technology assessment and SBD approaches that shift away from the static moral conceptions that are presently being considered and employed and instead towards an *imaginative* technology assessment and design methodology that begins with conceptual investigations built on the premises of an imaginative constitution.

The following section offers a preliminary example of how this reassembly of VSD, in particular, can be undertaken. Naturally, the limits of a single paper of this type restrict this praxis to only a cursory and speculative level. However, it should be the work of future research projects to explore how, if at all, the imaginative structure of the human mind, with its use of metonymy, metaphor, and narrative can be implemented in design strategies without subverting itself to strict principles.

5.2 A Praxis for Imaginative Value Sensitive Design (IVSD)

Because the emphasis of this paper is on the moral approach that VSD adopts, it leaves out a full discussion of VSD's restructuring in favour of emphasizing conceptual investigations alone.

At its core, VSD's conceptual investigations ideally begin with researchers exploring and determining various ‘value scenarios’ that can be envisioned and the potential tensions and insufficiencies that may arise (Friedman, Hendry, and Borning 2017). This is done by 1) determining the stakeholder population group and 2) beginning with a core value (e.g., safety, autonomy, privacy, etc.). This also involves determining the project values (sources of funding, industry interests, etc.) (Borning and Muller 2012).

An excellent way to begin rethinking the potential and nature of VSD is to:

²⁶ I have elsewhere criticised this Western moral dominance towards a preliminary course of adopting an approach of intersubjectivity (Umbrello 2018).

1. ~~Abandon the notion of a core set of root values. Instead of thinking of values as culturally instantiated, and objectively distilled, instead think of basic cognitive modules as universally shared.~~
2. ~~Similarly, the varied conception of ‘value’ as such must be explored as socioculturally situated, metaphorically conceptualized, narratively embodied, and developed through metonymy. VSD then must humble itself away from embedding objective values in design and towards a conception of design as embodying a dynamic value development. This is directly in line with VSD’s explicit goals of flexibility and reflexivity.~~

~~A means of approaching a salient IVSD could then be understood as injecting an imaginative tool-analysis alongside the basic VSD approach. Here this paper lays out the basic steps of conceptual investigations paired with a possible imaginative tool (iTool) that can be used to begin to rethink how VSD is practiced with some preliminary examples applied to autonomous vehicles:~~

Step 1: Determine project and designer values

iTool: Identify prototypical structures implicated in the design project

Step 2: Identify Direct and Indirect Stakeholders

iTool: Identify the plurality of moral traditions that these populations may be part of.

Application: Check to see if there are any *prima facie* moral overload between intra-population moral traditions, for example, balancing mobility, safety, and legality on the roadway.

Step 3: Determine Benefits and Harms for Stakeholders

iTool: Determine a definition of benefits/harms *a posteriori*, that, at least functionally, is acceptable to all the enrolled stakeholder communities.

Step 4: Identify Potential Values

iTool: Identify prototypical structures

Step 5: Distill Working Definition of Identified Values

iTool: Expand how socio-cultural prototypical values are instantiated from prototypes.

Application: how do users practice values of mobility and how does that conflict with other values such as safety and legality?

Step 6: Identify Value Conflict

iTool: Identify if the conflicts emerge from opposing values within a single moral tradition or because of cross-tradition incomparability?

Application: look for design requirements that satisfy opposing values that do not resort to rigid moral theories. Transparency in decision matrix algorithms need not be absolute, but dependent on other values. Hence, values such as transparency can be construed as a way to support or constrain other values, rather than an end in itself.

Step 7: Use Heuristic Tools in Stakeholder Value Elicitation

iTool: Identify if heuristic tools are trans-socially applicable. Ask if they aim at prototype structures or aprototypical expansions.

Application: When designers of AVs elicit stakeholder values they should be careful to be aware of their own biases (in the elicitation process itself) as well as employ debiasing tools that are aware of the prototype-aprototype structures of moral deliberation. The psychological literature on this is a good place to start.

6. Conclusion

What this paper has attempted to do is to provide an argument for why current design-for-values methodologies, particularly VSD, are insufficient for informing responsible innovation. The primary motivation for this is they rely exclusively on the MLTs of the Anglo-American tradition of philosophy to design technologies across cultures. The crux of the argument proposed has been to show how these MLTs do not accurately map on to innate human cognitive architecture that employs metaphor, narrative, metonymy, and imagination when engaging in moral deliberation. To this end, this paper has offered a way to incorporate an imaginative rationality in the VSD framework that may authentically map on to human cognition. Future research projects should empirically investigate which prototypical structures exist across socio-cultural divides and how those structures can be incorporated to form the core of VSD's conceptual investigations. In lieu of these empirical findings, the proposition of this paper remains conceptual and extrapolative. Yet, the observations from the cognitive sciences make a strong case for re-envisioning moral deliberations as an imaginative process nonetheless.

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